



Fig. 6: Mechanical egg base design using a bottle cap

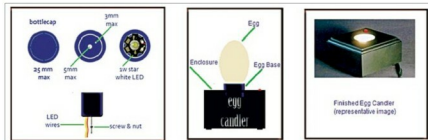


Fig. 7: Mechanical design outline



Fig. 8: Author's egg base for the egg candler

tor. A 1-kilo-ohm resistor (R2) limits the working current of LED1, while a 2.2-ohm resistor (R4) controls the current flowing through LED2. A 4.7-ohm resistor (R1) is used as the inrush current limiter, and a 390-kilo-ohm resistor (R3) across 1µF capacitor (C1) works as the bleeder resistor.

The transformerless power supply offers advantages of low cost and light weight. Also, it is reliable when operating with nominal load.

## Construction and testing

An actual-size PCB layout of the egg candler is shown in Fig. 4 and its components layout in Fig. 5. Link the battery to the star white LED (LED2) to generate the requisite light output. LED2 lights up when on/off switch (S1) is closed.

The egg base is made using a small bottle cap (23-25mm diameter) modified to suit the needs. A hole is made at the bottom of the cap for the wires of star white LED to pass through (Fig. 6). Besides, a small screw is fitted through another hole with washer and nut for the purpose of mounting later. Glue LED2 to the cap.

The finished egg base can be affixed to the top of a wooden/acrylic box with the circuit board (and battery) attached to the inner bottom of the enclosure. Refer Fig. 7 for mechanical design outline starting from the bottle cap to the proposed finished enclosure. The egg base prepared by the author is shown in Fig. 8.

After successful construction and initial testing, your home-made egg candler is ready for periodic candle-ding of eggs.

**Caution.** Beware, any experiment involving 230V AC mains voltage demands extreme care and absolute respect for electrical safety guidelines! **EFY**

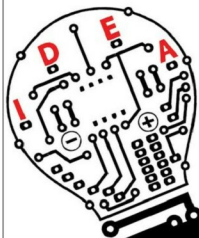


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